msvtx_el4hcal

overview · all regions and selections · rank 121 / 169


```
rank 121 / 169  by endcap / ht40 / D_equal

identical columns dropped in favour of this one:
    summedCaloClusters/el4hcal

region  selection D_equal D_pooled   125/15   125/55   200/50  400/100  600/150 1000/275   n_bkg
-------------------------------------------------------------------------------------------------
endcap  all         0.094    0.093    0.110    0.080    0.104    0.096    0.089    0.089  34,970
endcap  ht40        0.132    0.130    0.147    0.117    0.141    0.133    0.126    0.126   7,390
endcap  bkg split   0.048   below 27,580 / above 7,390

barrel  all         0.166    0.159    0.151    0.202    0.181    0.172    0.160    0.139   1,089
barrel  ht40        0.306    0.299    0.286    0.340    0.321    0.312    0.299    0.279     220
barrel  bkg split   0.178   below 869 / above 220

all     all         0.089    0.085    0.097    0.090    0.099    0.092    0.081    0.075  36,059
all     ht40        0.129    0.125    0.137    0.130    0.139    0.132    0.121    0.115   7,610
all     bkg split   0.051   below 28,449 / above 7,610

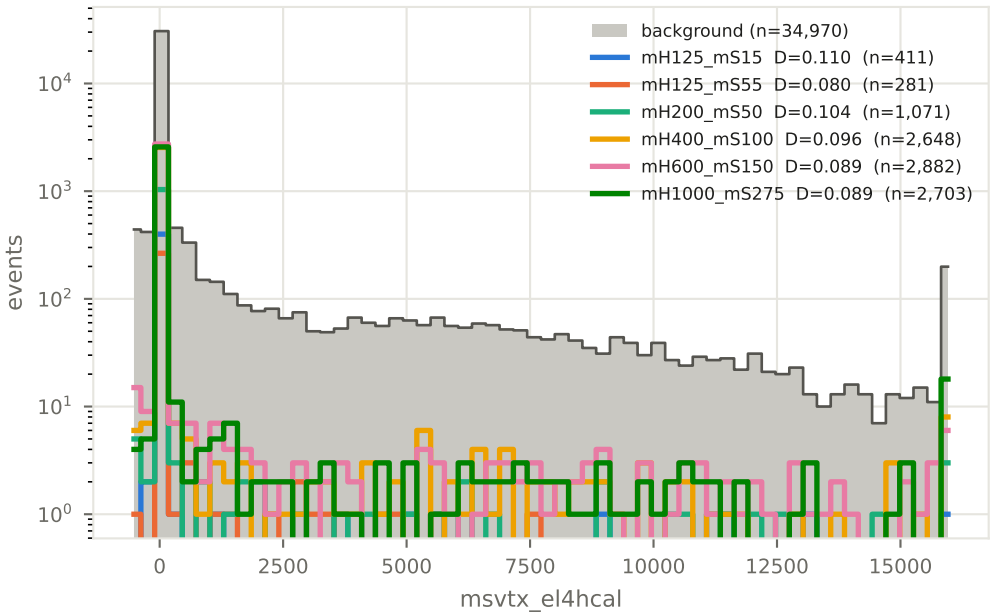
selections: all = every event;  ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2
```

msvtx_el4hcal

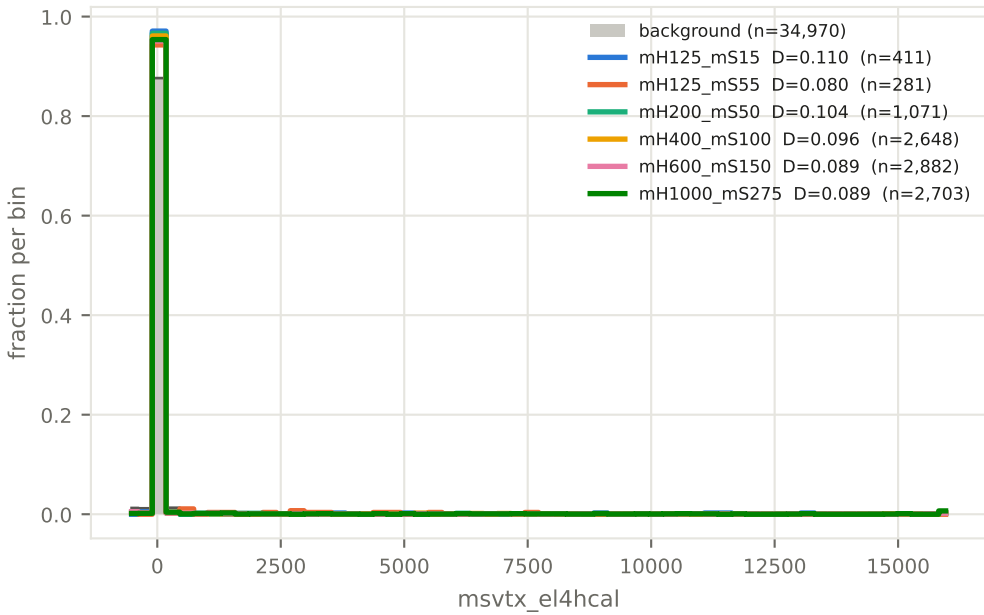
region: ENDCAP · selection: all (every event) · rank 121 / 169

D_equal 0.094 D_pooled 0.093 min 0.080 (mH125_mS55) max 0.110 (mH125_mS15) bkg n = 34,970 sig n = 9,996

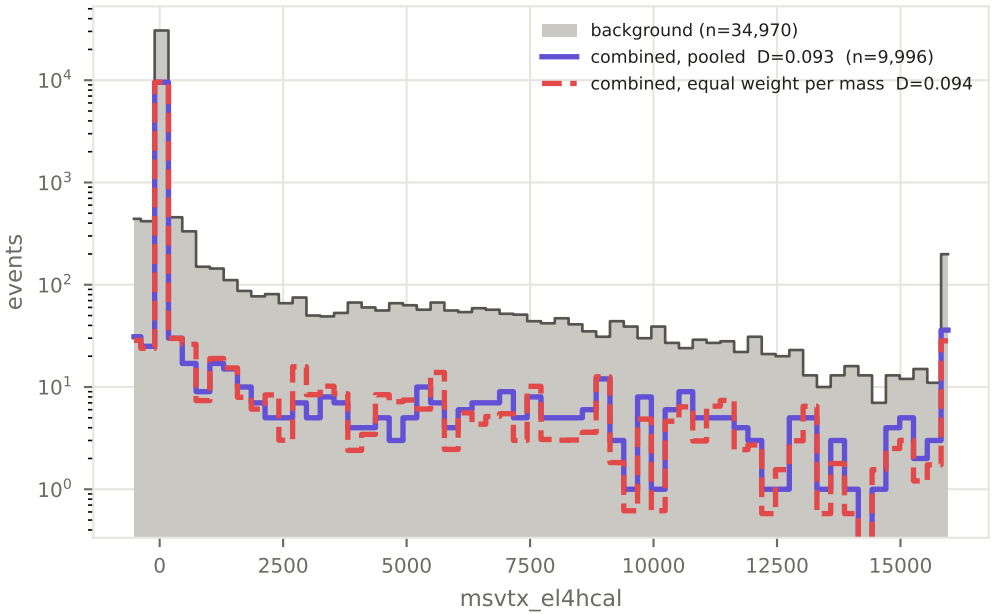
each mass point vs background · raw



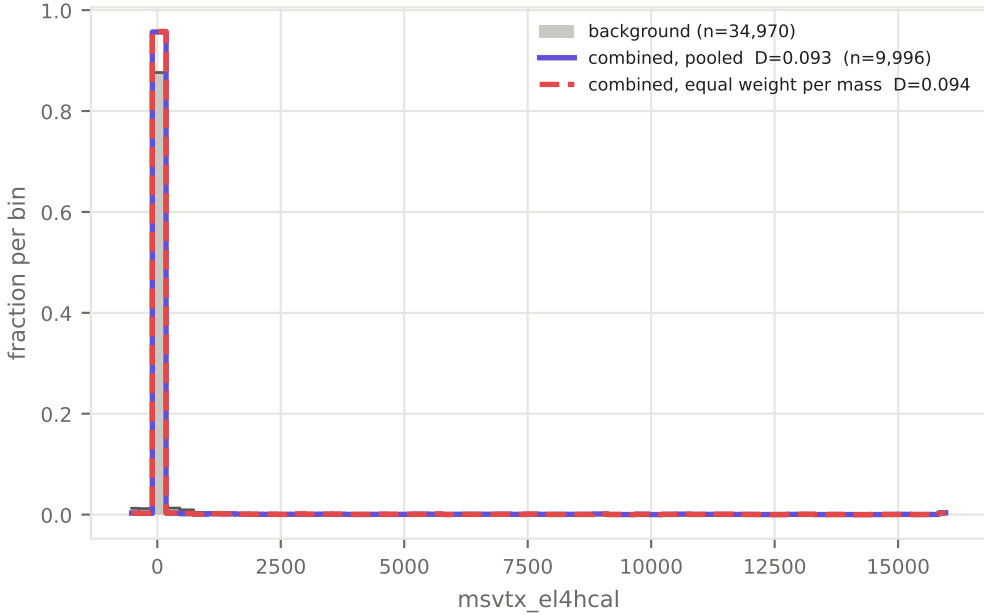
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



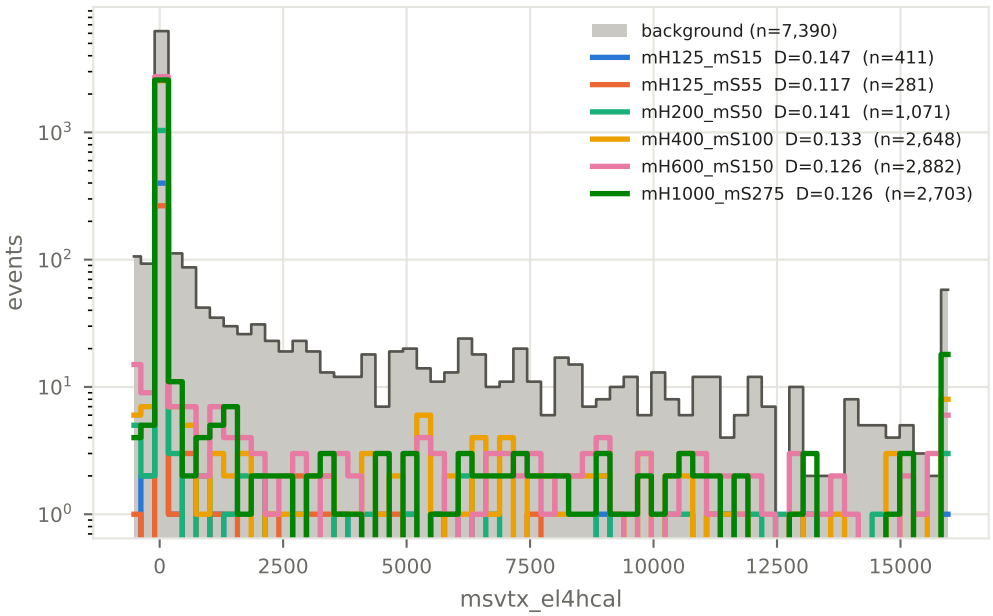
edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

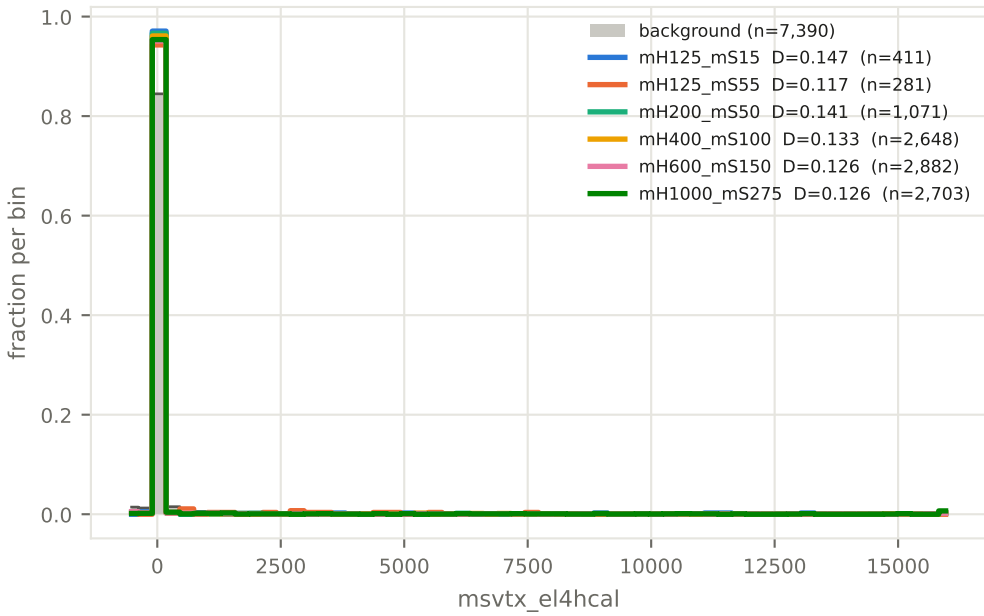
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D_equal 0.132 D_pooled 0.130 min 0.117 (mH125_mS55) max 0.147 (mH125_mS15) bkg n = 7,390 sig n = 9,996

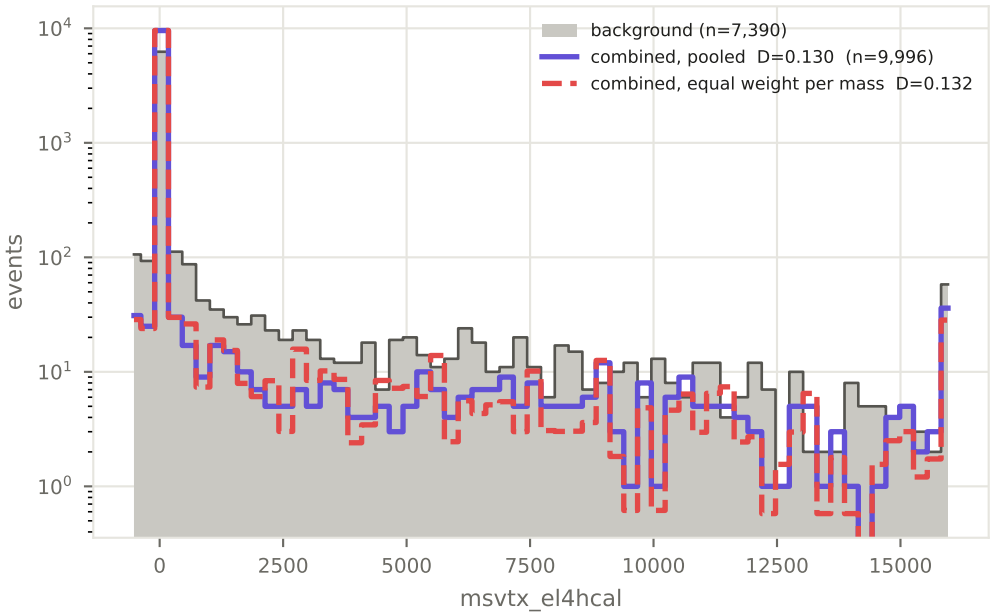
each mass point vs background · raw



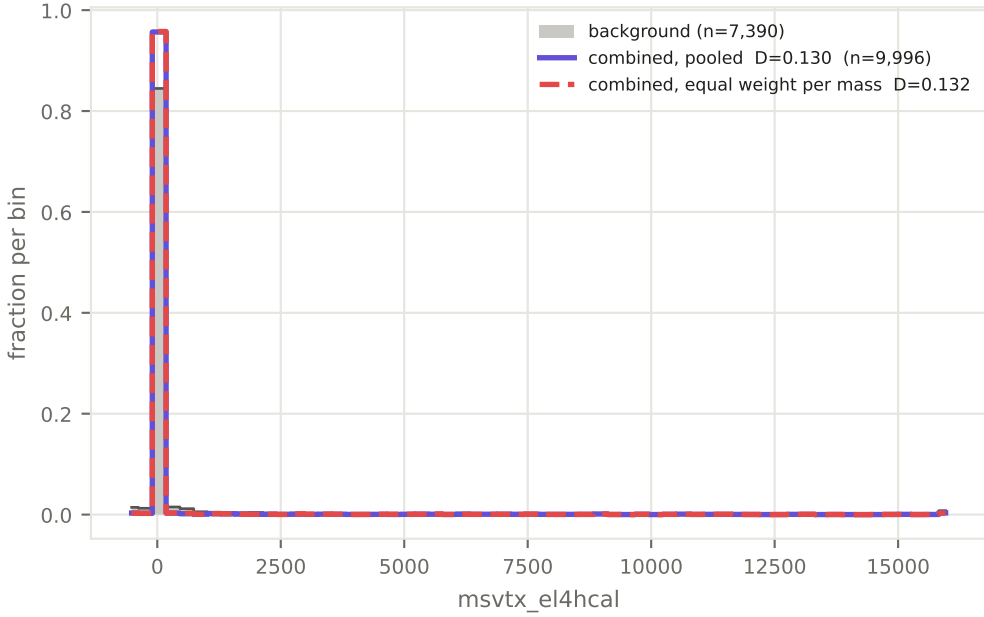
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

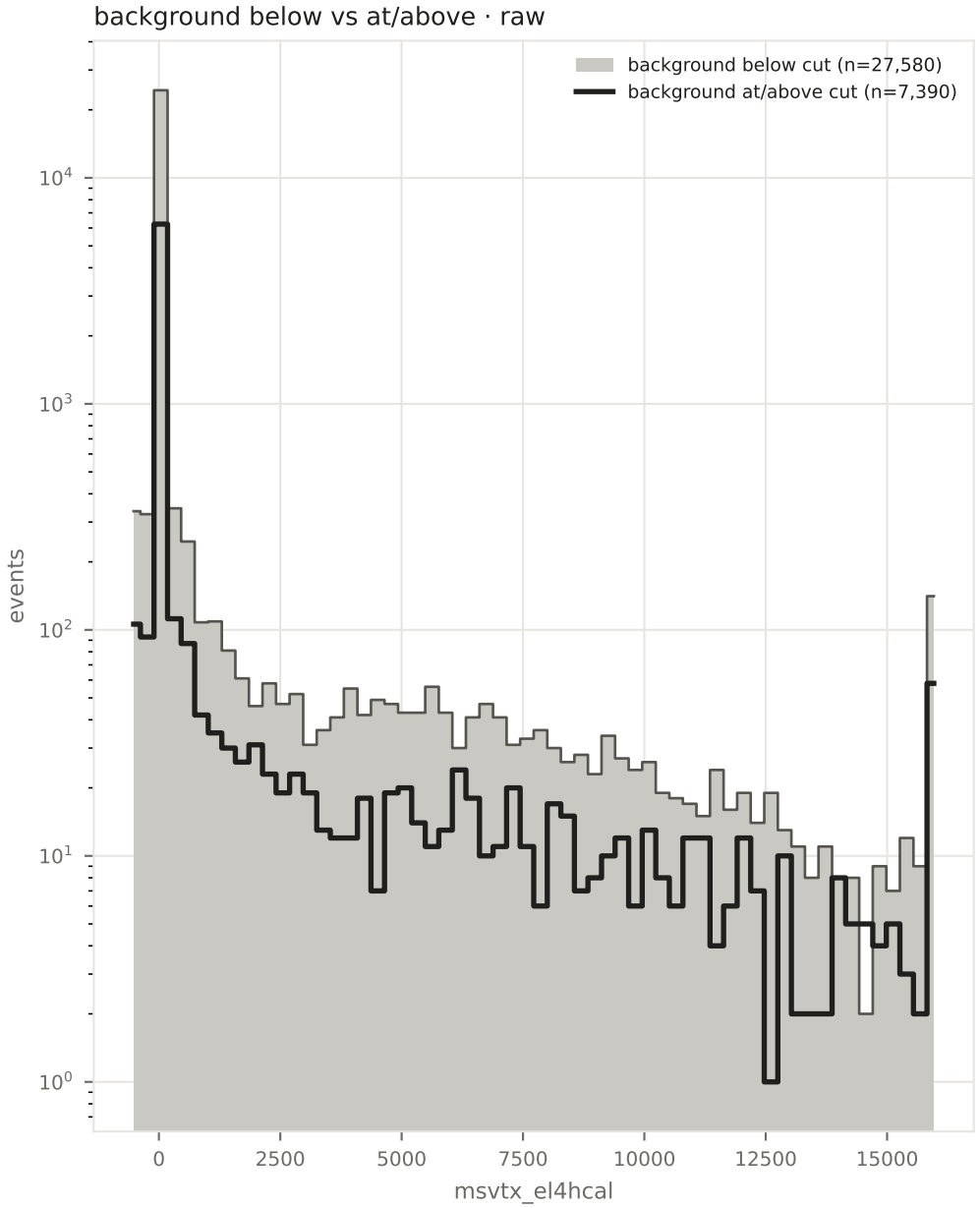


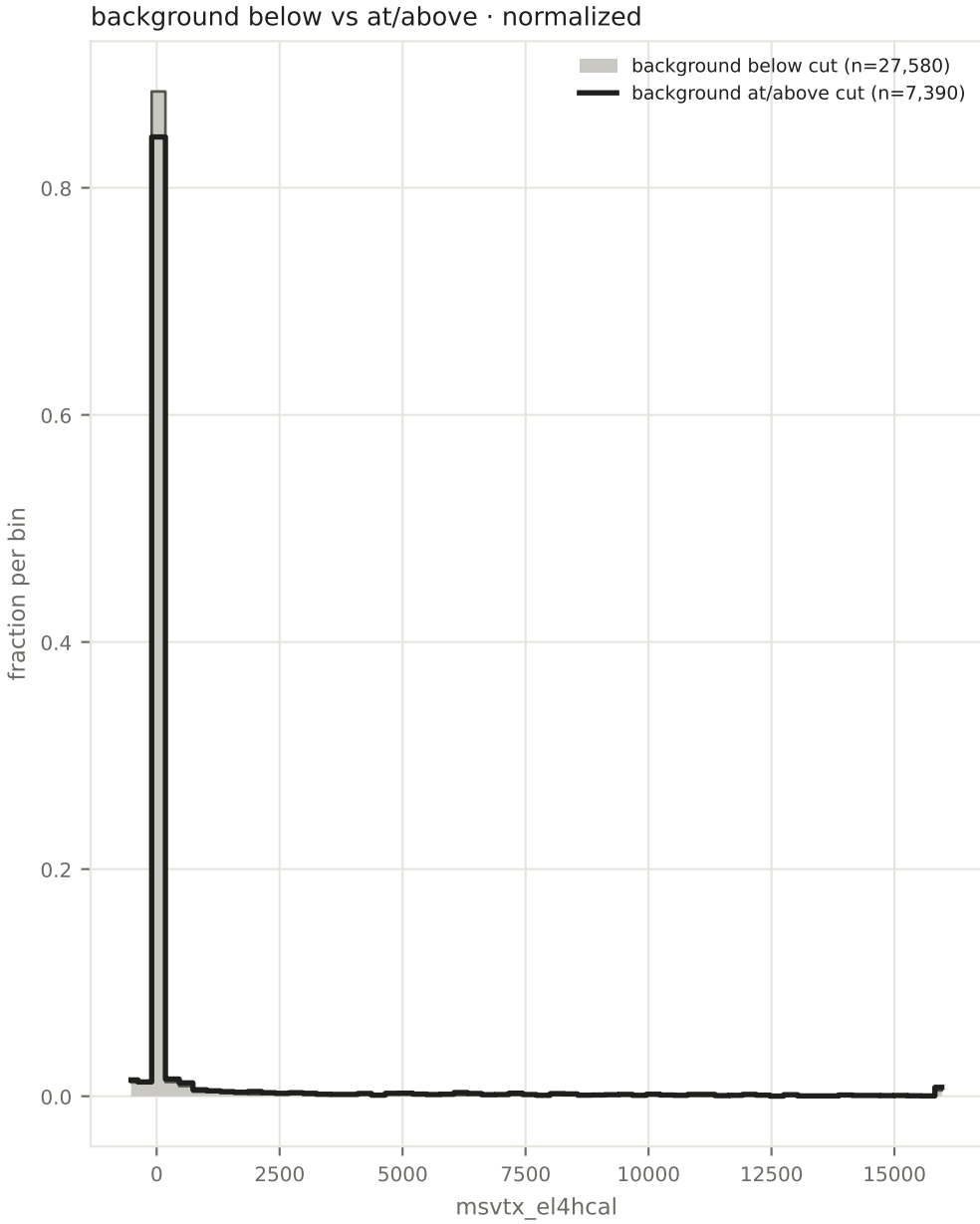
edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D(below vs above) 0.048 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



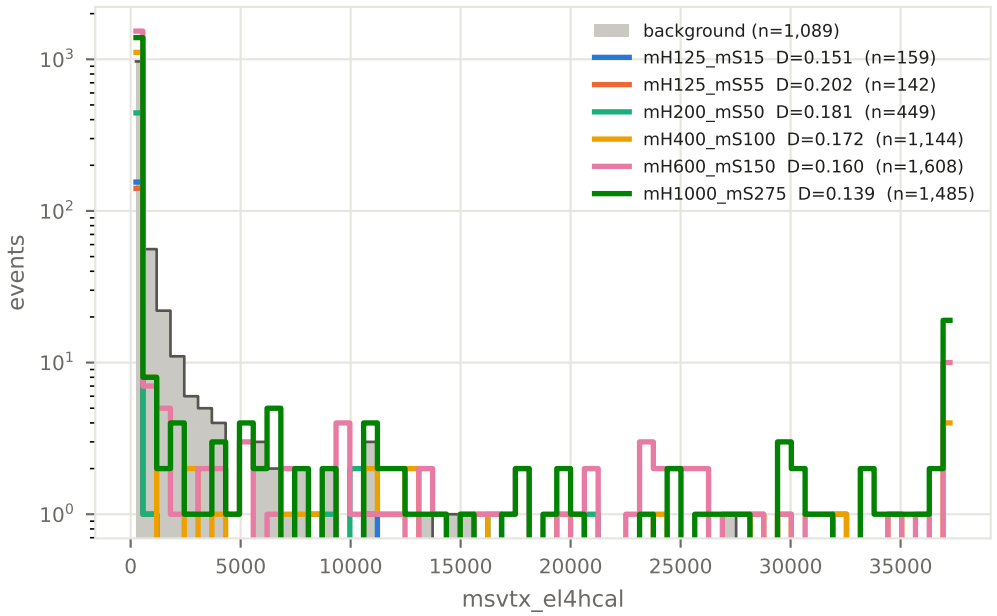


msvtx_el4hcal

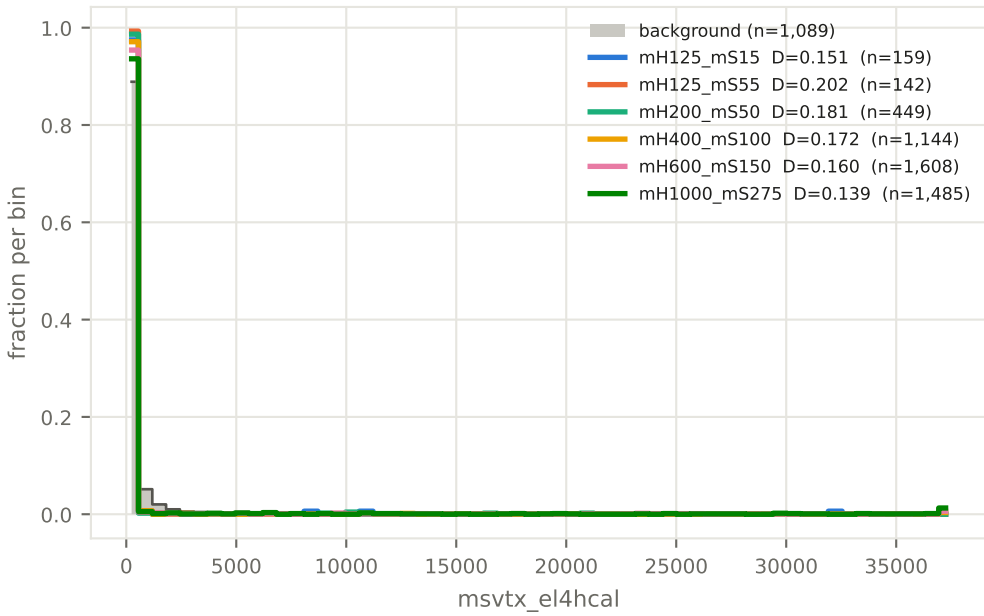
region: BARREL · selection: all (every event) · rank 121 / 169

D_equal 0.166 D_pooled 0.159 min 0.139 (mH1000_mS275) max 0.202 (mH125_mS55) bkg n = 1,089 sig n = 4,987

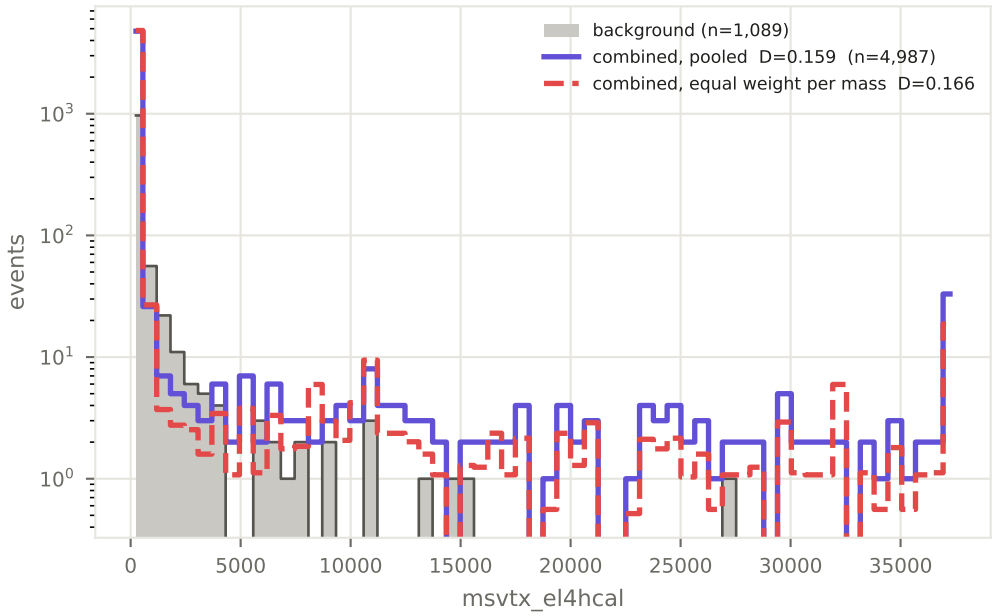
each mass point vs background · raw



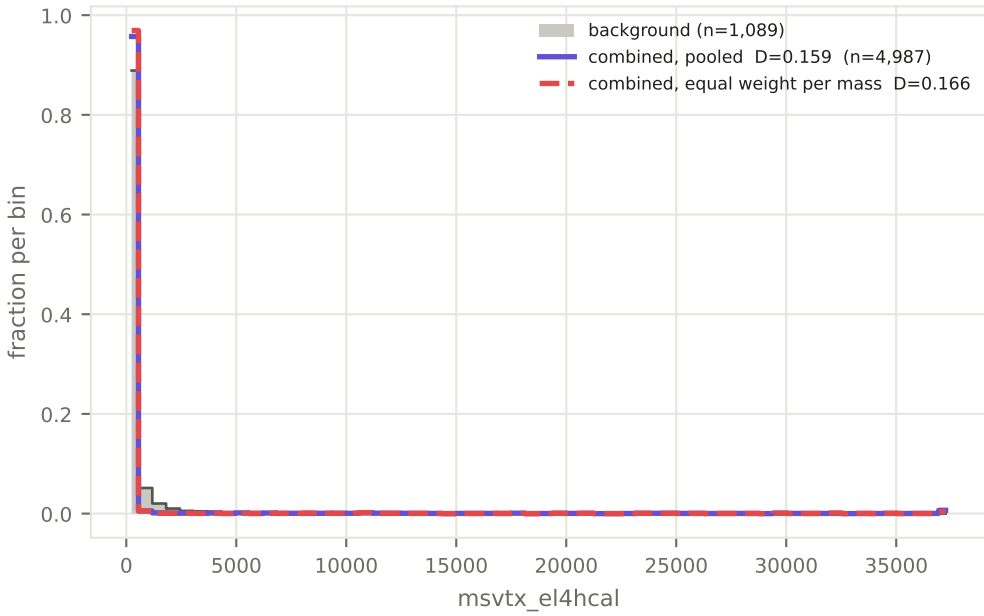
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



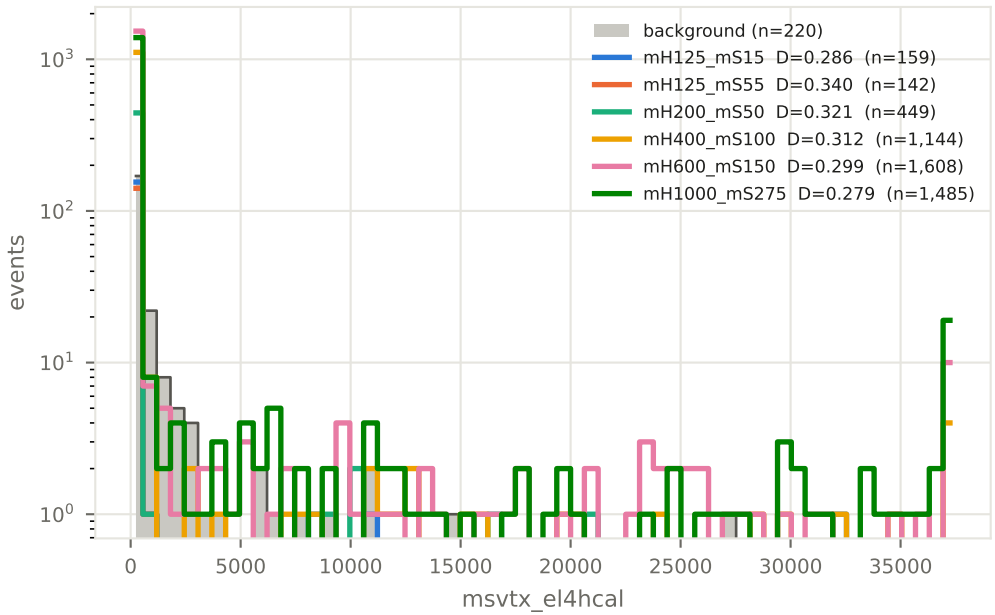
edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

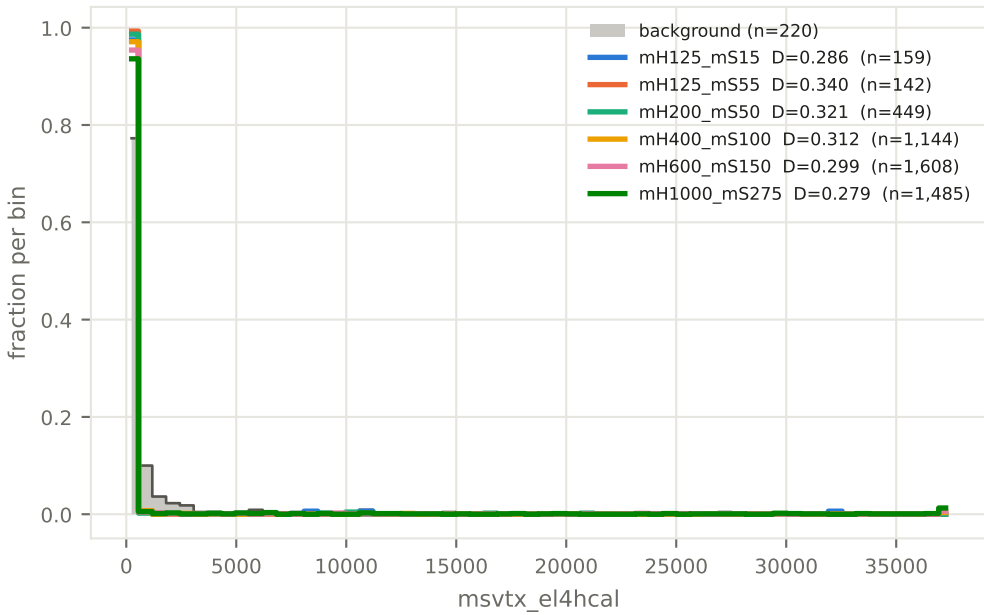
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D_equal 0.306 D_pooled 0.299 min 0.279 (mH1000_mS275) max 0.340 (mH125_mS55) bkg n = 220 sig n = 4,987

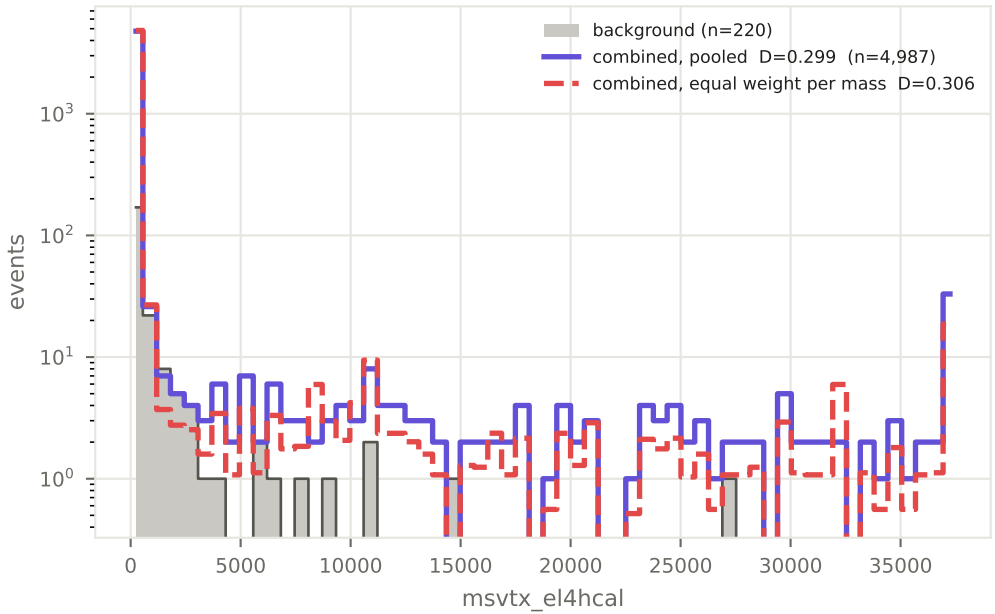
each mass point vs background · raw



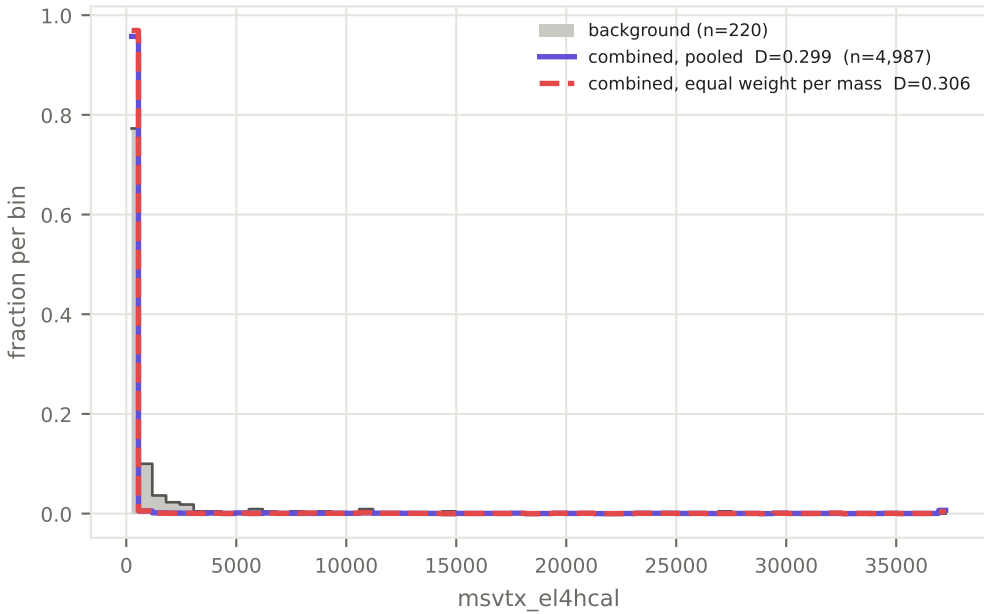
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

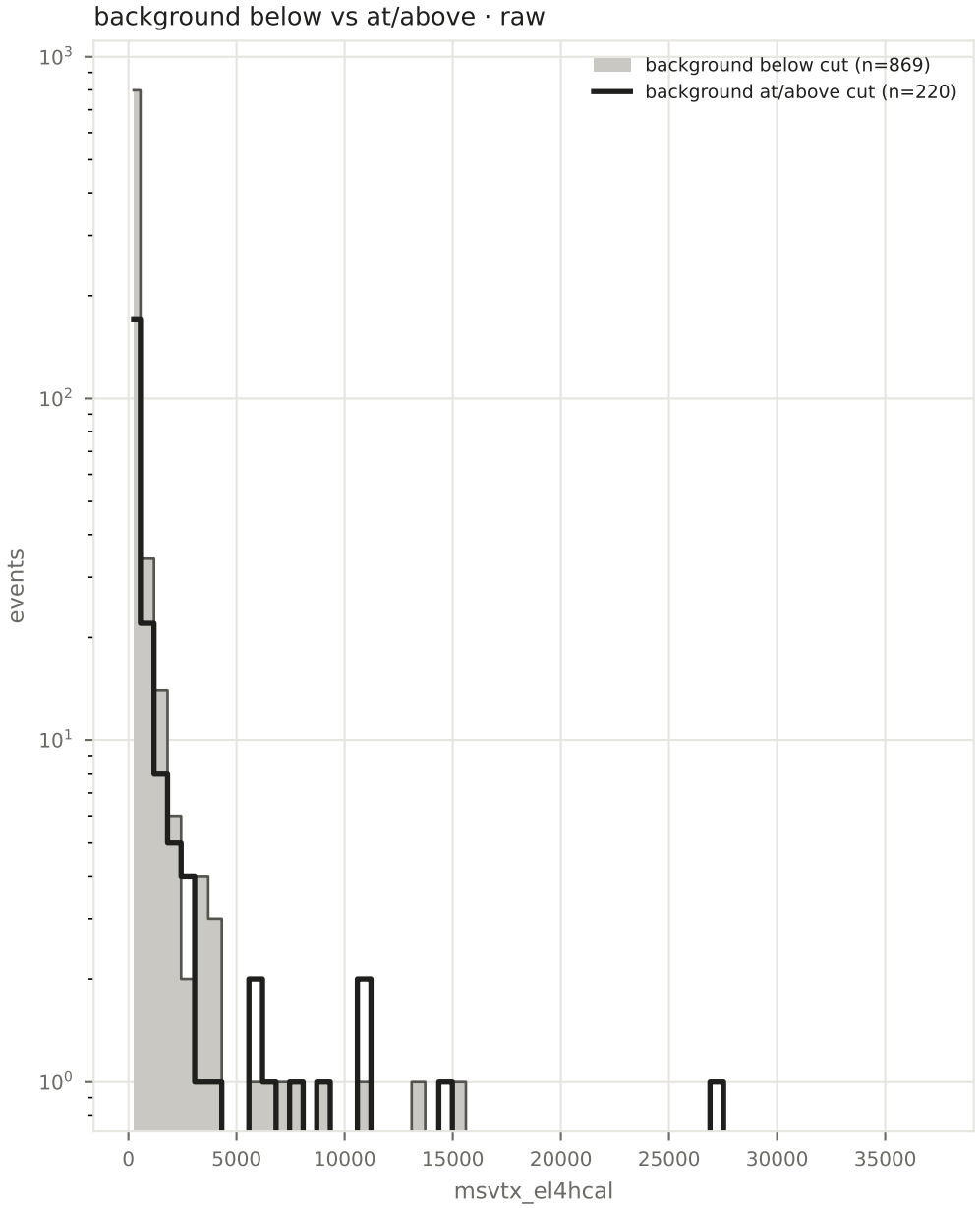


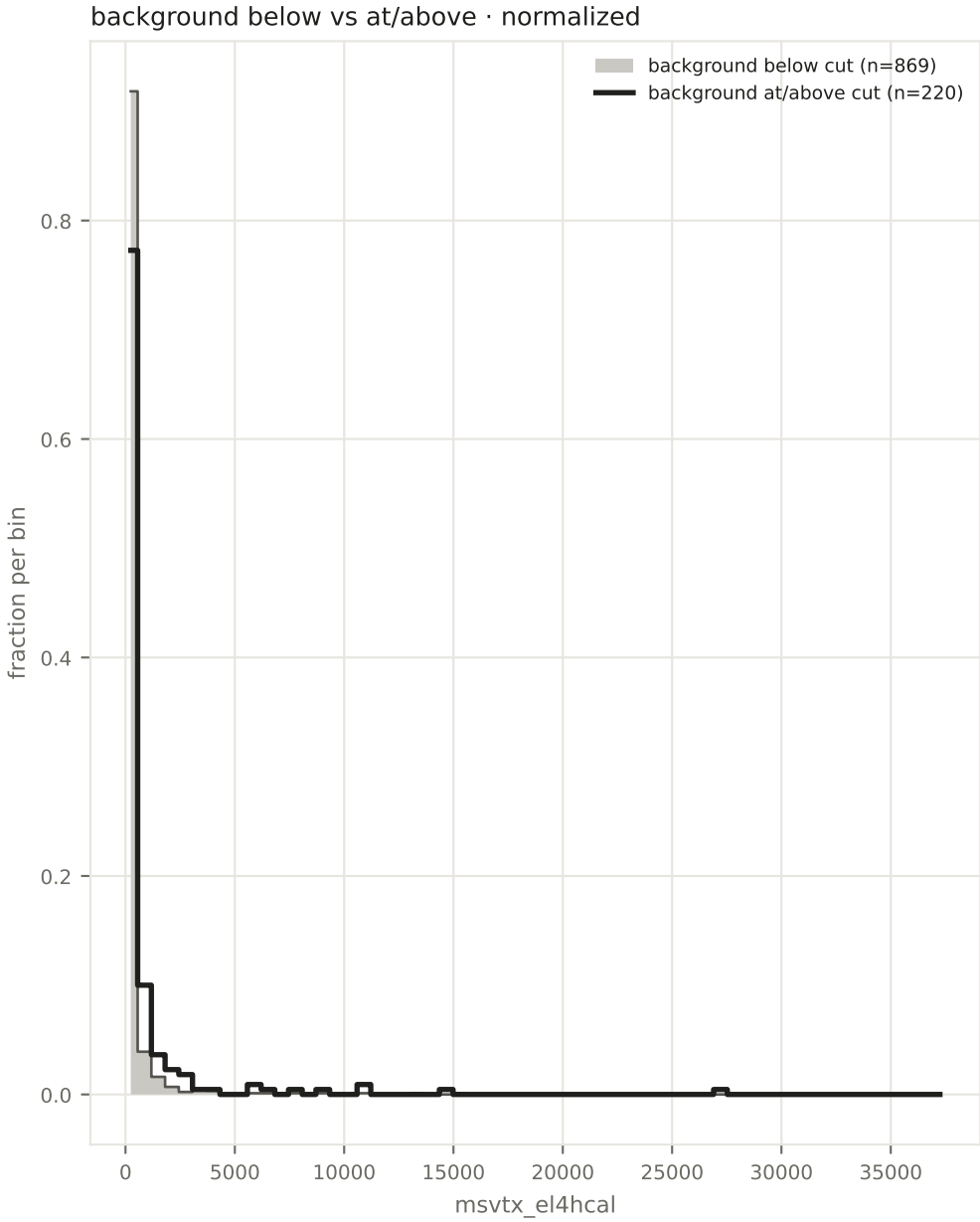
edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D(below vs above) 0.178 n below = 869 n above = 220 large D = background shape changes across the cut



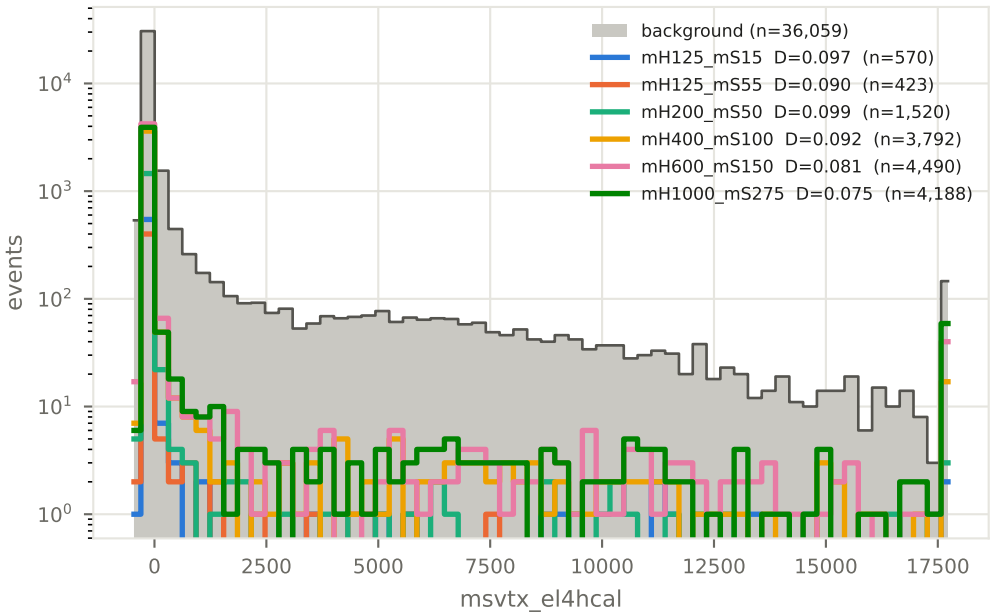


msvtx_el4hcal

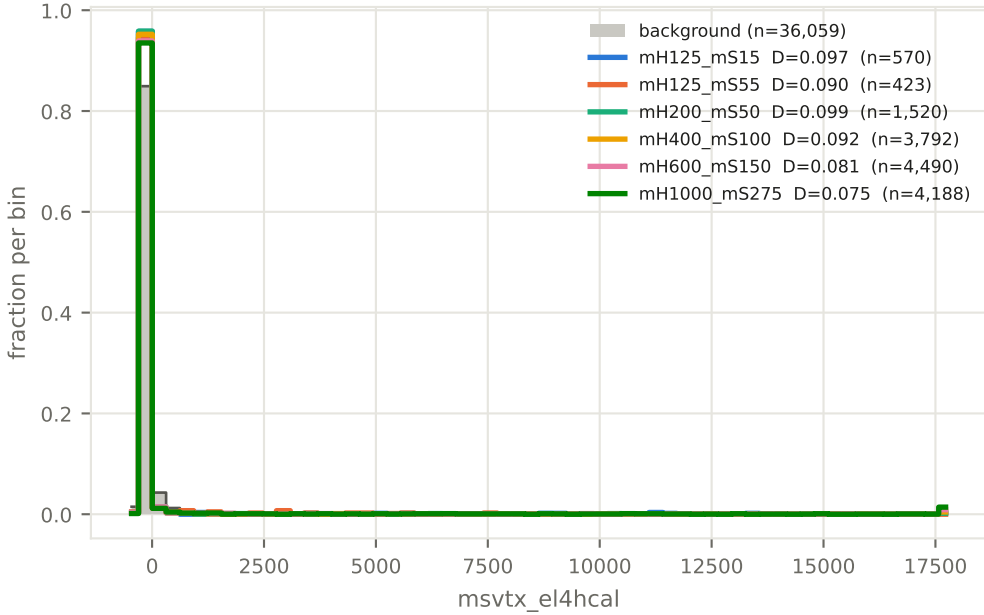
region: ALL · selection: all (every event) · rank 121 / 169

D_equal 0.089 D_pooled 0.085 min 0.075 (mH1000_mS275) max 0.099 (mH200_mS50) bkg n = 36,059 sig n = 14,983

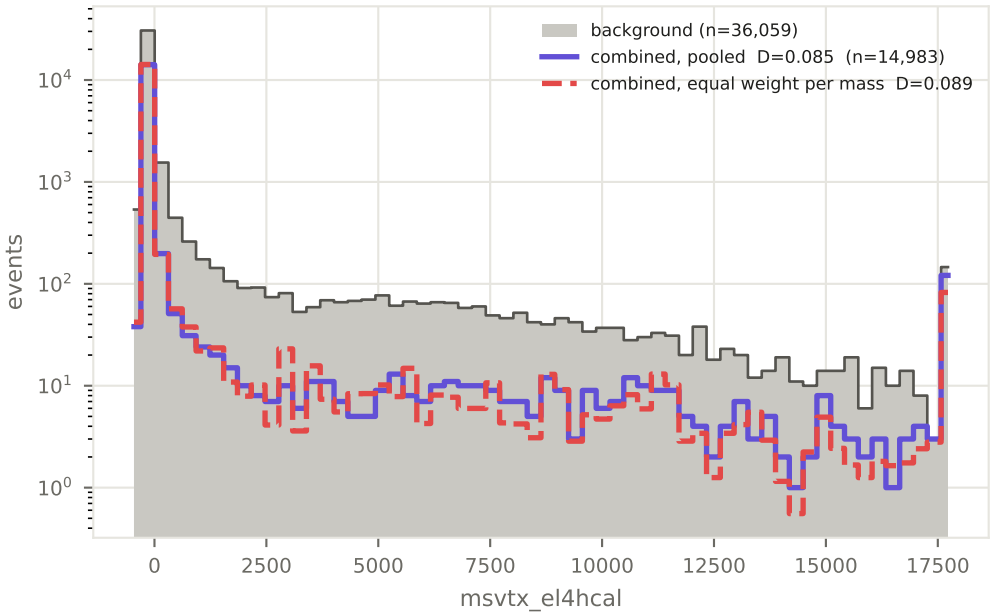
each mass point vs background · raw



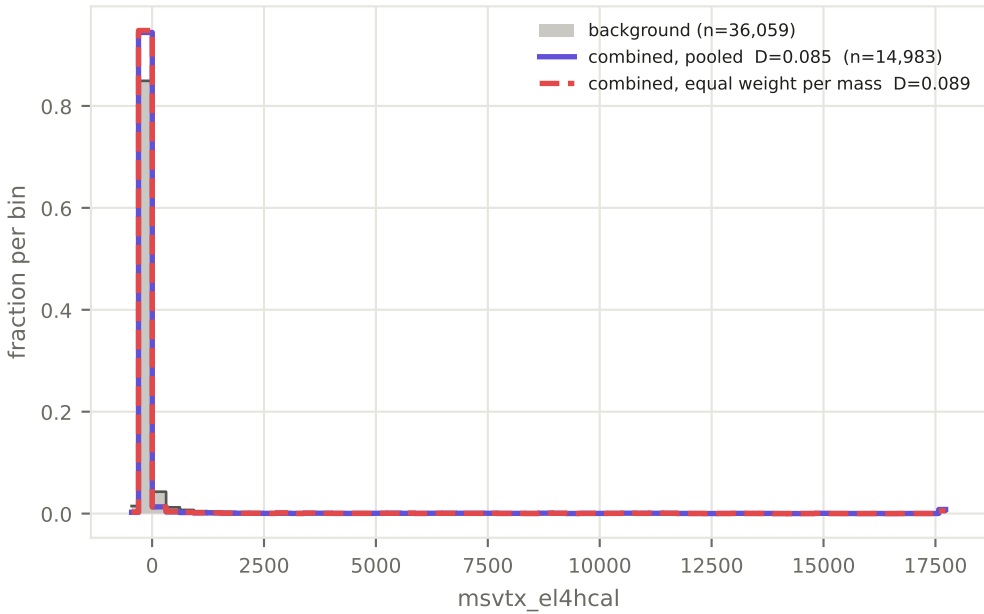
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



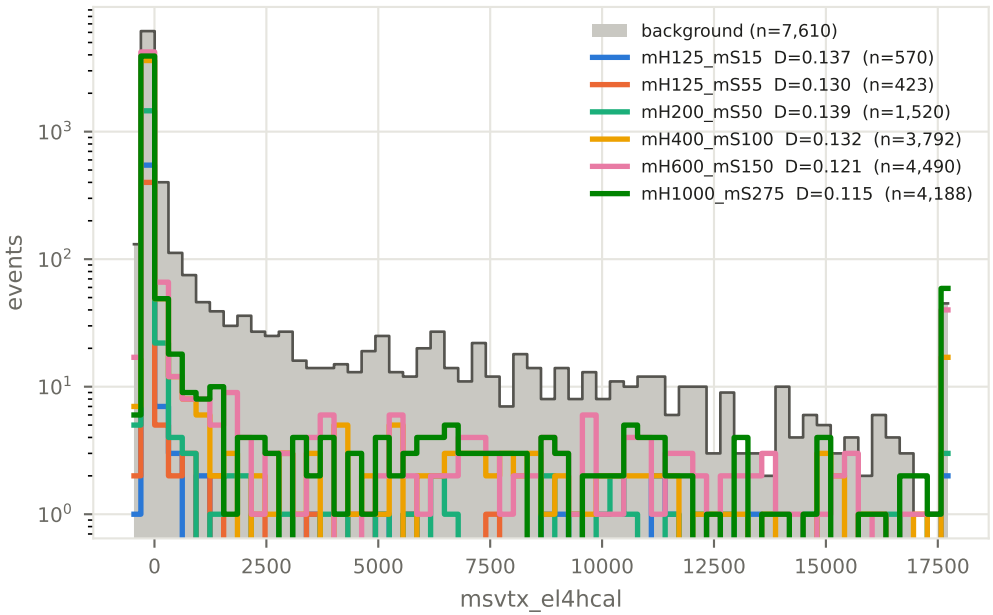
edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

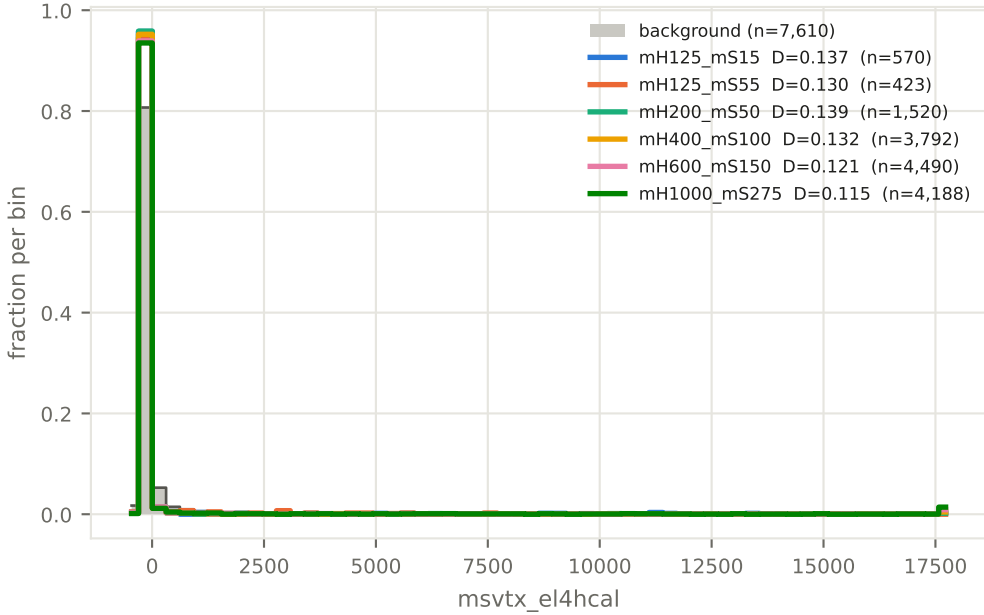
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D_equal 0.129 D_pooled 0.125 min 0.115 (mH1000_mS275) max 0.139 (mH200_mS50) bkg n = 7,610 sig n = 14,983

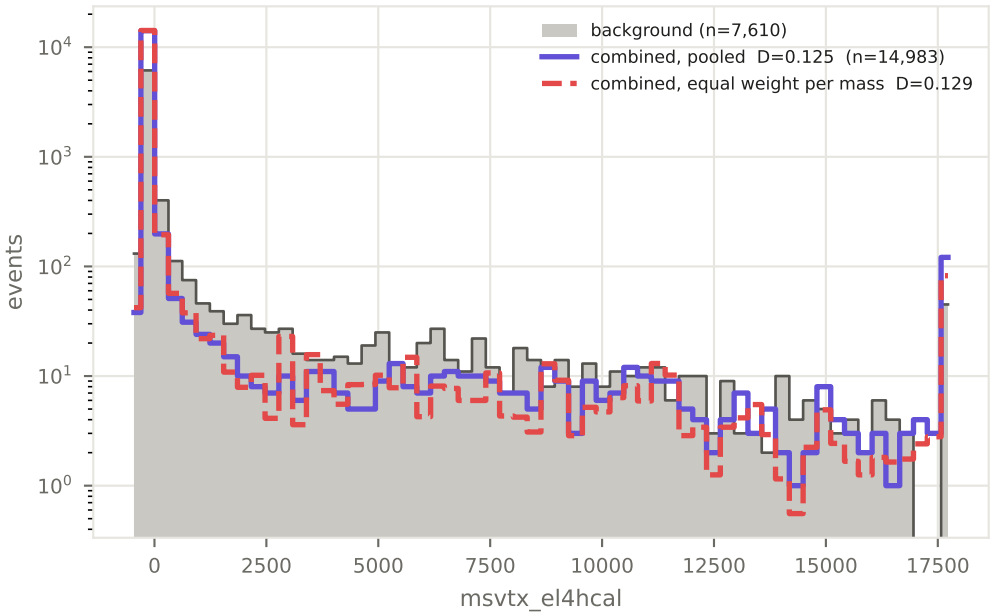
each mass point vs background · raw



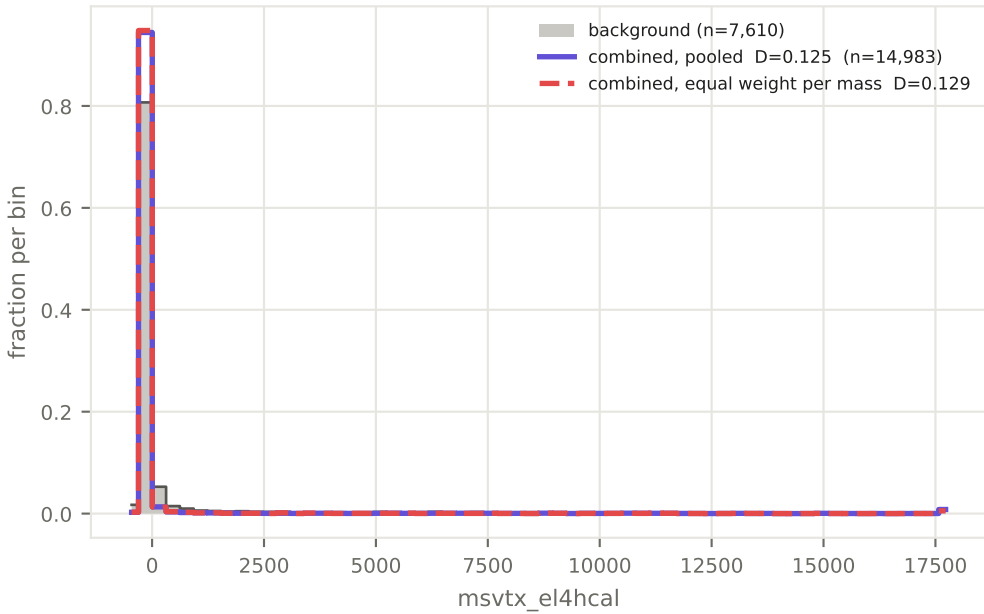
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

msvtx_el4hcal

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 121 / 169

D(below vs above) 0.051 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

